



GENERAL APTITUDE

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Interest

If P = Principal, R = Rate of interest, N = Time in years, I = Interest, A = Amount

Then **$A = P + I$**

Simple Interest

$$\text{S.I.} = (P \times R \times N) / 100$$

Basic principal remains constant.

S.I. is good example of AP(Arithmetic Progression)

Compound Interest

$$A = P (1 + R/100)^T$$

$$\text{C.I.} = A - P$$

T = periods of compounding,

R = rate for compounding period

Basic principal keeps on increasing as we get interest on interest.

C.I. is good example of GP(Geometric Progression)



Interest

Q. A shopkeeper with an OD facility at 18% with a bank borrowed Rs. 15000 on Jan 8, 2011 and returned the money on June 3, 2011 so as to clear the debt. The amount that he paid was -

- A. Rs. 16080 B. Rs. 16280 C. Rs. 16400 D. None of these

Soln:

- $P = 15000$, $r = 18\%$,
- $T = 23(\text{jan}) + 28(\text{feb Nonleap}) + 31(\text{march}) + 30(\text{April}) + 31(\text{may}) + 3(\text{june}) = 146$ days
- $146/365$ days = $2/5$ years.
- $SI = 15000 \times 18 \times 2/5 \times 1/100 = 30 \times 18 \times 2 = 1080$

$$\begin{aligned}\text{Amount} &= P + SI \\ &= 15000 + 1080 \\ &= \text{Rs. } 16080\end{aligned}$$

Ans: A



Interest

Q. A farmer borrowed Rs.3600 at 15% simple interest per annum. At the end of 4 years, he cleared this account by paying Rs.4000 and a donkey. The cost of the donkey is -

A. Rs. 1000

B. Rs. 1200

C. Rs. 1550

D. Rs. 1760

Soln:

SI for 4 years = Rs. $(3600 \times 0.15 \times 4) = \text{Rs. } 2160$

Amount after 4 years = Rs. $(3600 + 2160) = \text{Rs. } 5760$

Cost of donkey = Rs. $(5760 - 4000) = \text{Rs. } 1760$

Ans: D



Interest

Q. In 4 years, Rs. 6500 amounts to Rs. 8840 at a certain rate of interest. In what time will Rs.1600 amounts to Rs.1816 at the same rate?

- A. 3years B. 1.5years C. 2years D. 2.5years

Ans: B



Interest

Q. P =Rs. 2000, R =10%, N =2yrs , Find A and CI

Soln:

$$A = 2000\left(1 + \frac{10}{100}\right)^2$$

$$= 2000\left(\frac{110}{100}\right)^2$$

$$= 2000\left(\frac{121}{100}\right)$$

$$= \text{Rs. } 2420$$

$$\text{CI} = 2420 - 2000 = \text{Rs. } 420$$

$$2000 \rightarrow 10\% = 200$$

$$10\% \quad 10\%$$

$$2000 \longrightarrow 2200 \longrightarrow 2420$$

$$\text{CI} = 2420 - 2000 = 420$$



Interest

Q. Simple interest on a certain sum of money for 3 years at 8% per annum is half the compound interest on Rs. 4000 for 2 years at 10% per annum. The sum placed on simple interest is:

A. Rs. 1550

B. Rs. 1650

C. Rs. 1750

D. Rs. 2000

Soln:

$$A = P \left(1 + \frac{R}{100} \right)^N = 4000 \left(1 + \frac{10}{100} \right)^2 = 4000 \times \left(\frac{11}{10} \right)^2 = 4000 \times \frac{11}{10} \times \frac{11}{10} = \text{Rs. } 4840$$

OR

$$4000 \xrightarrow[1^{\text{st}} \text{ yr}]{10\%} 4400 \xrightarrow[2^{\text{nd}} \text{ yr}]{10\%} 4840$$

$$CI = A - P$$

$$CI = 4840 - 4000 = \text{Rs. } 840$$

Ans: C

$$SI = \frac{1}{2} CI$$

$$\frac{PNR}{100} = \frac{1}{2} \times 840$$

$$\frac{P \times 3 \times 8}{100} = 420$$

$$P(\text{sum}) = \frac{420 \times 100}{3 \times 8} = \text{Rs. } 1750$$



Interest

Q. P =Rs. 4000, R =20% per annum, N =6months.Find CI computed quarterly for given period.

Soln:

N =6months(2 quarterly)

rate(R) = 20 % per annum = 5 % quarterly

After every 3 months CI will be calculated.

	by $5\%=200$	by $5\%=210$	
4000	→	4200	→
			4410

I = 4410 -4000

= Rs. 410



Interest

Q. Difference between Compound interest & simple interest on a sum placed at 8% p.a. compounded annually for 2 years is Rs 128. Find the Principal

- A. 20000 B. 24000 C. 26000 D. 15000

• **Soln:**

- Let the principal be P = Rs. 100.
- time N = 2 years, rate of interest R = 8% per annum
- simple interest = $\frac{PNR}{100} = \frac{100 \times 8 \times 2}{100} = \text{Rs. } 16$

- CI (for 2 years)

- 8% 8%
- 100 $\xrightarrow{\quad}$ 108 $\xrightarrow{\quad}$ 116.64

	16.64		
P	SI	CI	Diff
100	16	16.64	0.64

- 0.64 -> 100
- 128 -> ?
- $\frac{12800}{0.64} = \text{Rs. } 20000$



Interest

Q. Difference between Compound interest & simple interest on a sum placed at 8% p.a. compounded annually for 2 years is Rs 128. Find the principal

- A. 20000
- B. 24000
- C. 26000
- D. 15000

• **Soln:**

- Let the principal be $P = \text{Rs. } 100$.
- time $N = 2$ years, rate of interest $R = 8\%$ per annum
- simple interest $= \frac{PNR}{100} = \frac{100 \times 8 \times 2}{100} = \text{Rs. } 16$
- compound amount $= P(1 + \frac{R}{100})^N$
- $= 100 \times (1 + \frac{8}{100})^2 = 100 \times (\frac{108}{100})^2 = 100 \times (\frac{11664}{10000}) = \frac{11664}{100} = 116.64$
- compound interest $= \text{compound amount} - \text{principal}$
- $C.I = A - P$
 $= 116.64 - 100 = \text{Rs. } 16.64$
- the difference between the compound interest and simple interest $= 16.64 - 16.00 = \text{Rs. } 0.64$
- $\frac{0.64}{100} \rightarrow 100$
- $\frac{128}{0.64} \rightarrow ?$
- $= \frac{128 \times 100}{0.64} = 20000$
- Thus, the principal is Rs. 20000.



Ans : A

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Interest

- If the difference between compound and simple interest is of **two years** than,
Difference = $P(R)^2/(100)^2$
Where P = principal amount, R = rate of interest
- If the difference between compound and simple interest is of **three years** than,
Difference = $3 \times P(R)^2/(100)^2 + P (R/100)^3$.
Here also, P = principal amount, R = rate of interest



Interest(Assignment)

Q. The difference between simple interest and compound interest on a certain sum of money for 2 years at 4% per annum is Rs 20. Find the sum.

A. 12500

B. 25000

C. 12000

D. 17500

Ans: A



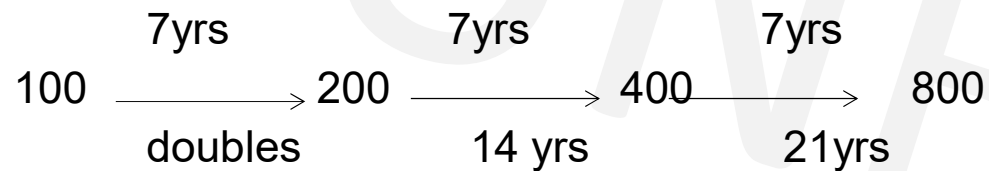
Interest(Assignment)

Q. A sum of money placed at compound interest doubles in 7 years. In how many years the principal becomes-

- a. 4 times of itself
- b. 8 times of itself

Soln:

Let initial value be 100



- a. In 14yrs
- b. In 21 yrs

OR

100----->200 in 7 years
200----->400 in again 7 years then,
400----->800 in 7 years again, thus
the time becomes= 7+7+7= 21 years.



Interest(Assignment)

Q. When annual compounding is done, a sum amounts to Rs 5000 in 6 years and 7200 in 8 years.
What is the int rate?

- A. 10% B. 15% C. 20% D. 25%

Soln

Let P be the principal & R the int rate

$$\rightarrow 5000 = P(1+R/100)^6 \dots\dots(1)$$

$$\rightarrow 7200 = P(1+R/100)^8 \dots\dots(2)$$

$$\rightarrow 36/25 = (1+R/100)^2$$

$\rightarrow$ Taking square roots of both sides

$$\rightarrow 1+R/100 = 6/5$$

$$\rightarrow R/100 = 1/5$$

$$\rightarrow R = 20\%$$

Ans: C



Interest(Assignment)

Q. A sum fetched a total simple interest of Rs.7056 at the rate of 8 percent per year in 7 years. What is the sum?

A. Rs 12600

B) Rs 15120

C) Rs 10080

D) Rs 7560

Ans : A



Interest(Assignment)

Q. Find the compound interest on Rs. 15,625 for 9 months at 16% per annum compounded quarterly.

A. Rs. 1851

B. Rs. 1941

C. Rs. 1951

D. Rs. 1961

Ans: C



Interest(Assignment)

Q. What is the difference between the simple interest on a principal of Rs. 500 being calculated at 5% per annum for 3 years and 4% per annum for 4 years?

A.Rs. 5 B.Rs. 10 C.Rs. 20 D.Rs. 40 E. None of these

$$\begin{aligned} SI_1 &= P N_1 R_1 / 100 \\ &= \frac{500 \times 3 \times 5}{100} = \text{Rs. } 75 \end{aligned}$$

$$\begin{aligned} SI_2 &= P N_2 R_2 / 100 \\ &= \frac{500 \times 4 \times 4}{100} = \text{Rs. } 80 \end{aligned}$$

$$\text{Difference} = 80 - 75 = \text{Rs. } 5$$

OR

$$500 \Rightarrow 15\% \uparrow \Rightarrow 575 \text{ (1st case)}$$

$$500 \Rightarrow 16\% \uparrow \Rightarrow 580 \text{ (2nd case)}$$

$$\text{difference} = 580 - 575 = \text{Rs. } 5$$

Ans : A



Interest(Assignment)

Q. A sum of money placed at compound interest doubles itself in 4 years. In how many years will it amount to 8 times?

- A. 9 years B. 8 years C. 27 years D. 12 years

Ans: D



Interest(Assignment)

Q. Difference between Compound interest & simple interest on a sum placed at 20% per annum compounded annually for 2 years is Rs. 72. Find the sum.

- A. Rs. 2400 B. Rs. 8400 C. Rs. 1800 D. Rs. 900

Ans : C



Interest(Assignment)

Q. What is the simple interest on a sum of Rs. 700 if the rate of interest for the first 3 years is 8% per annum and for the last 2 years is 7.5% per annum?

A.Rs. 269.5 B.Rs. 283 C.Rs. 273 D.Rs. 280 E. None of these

Ans: C



Interest(Assignment)

Q. Rs.2100 is lent at compound interest of 5% per annum for 2 years. Find the amount after two years.

- A.Rs. 2300 B.Rs. 2315.25 C.Rs. 2310 D.Rs. 2320 E. None of these

• **Soln:**

• $A = P (1 + R/100)^T$

• $A = 2100(1+5/100)^2$

• $A = 2100 \times [105/100]^2$

• $A = \frac{2100 \times 11025}{10000}$

• Amount, A=Rs.2315.25

• **Ans : B**



Interest(Assignment)

Q. A man borrowed total Rs 2500 at Simple interest from two money lenders. He paid interest at 12% p.a. to one and 14% p.a. to the other. The total interest paid for the year was Rs.326. How much did he borrow at 14%?

A. Rs 1000

B. Rs 1200

C. Rs 1300

D. Rs 1500

Soln:

Let, x = Principal at 12%

&

2500-x = Principal at 14%

$$\text{SI at Rs.}x = \frac{x \times 1 \times 12}{100} = \frac{12x}{100} = \frac{3x}{25}$$

$$\text{SI at Rs.}2500 - x = \frac{2500 - x \times 1 \times 14}{100} = \frac{(2500 - x) \times 7}{50} = \frac{17500x - 7x}{50}$$

$$\text{SI at } x + \text{SI at } 2500 - x = 326$$

Substitute and solving the equation gives x =Rs. 1200

We need Principal at 2500-x = 2500 – 1200 =Rs. 1300

Ans: C



Interest(Assignment)

Q.A certain sum of money amounts to Rs. 704 in two years and Rs 800 in 5 years. Find the Principal.

A. Rs. 640

B. Rs. 600

C. Rs. 550

D. Rs. 450

Ans: A



Interest(Assignment)

Q. What should be the simple interest obtained on an amount of Rs 5,760 at the rate of 6% p.a. after 3 years?

- A. Rs 1036.80
- B. Rs 1666.80
- C. Rs 1336.80
- D. Rs 1063.80
- E. None of these

Ans : A



Interest(Assignment)

Q. The simple interest on Rs.1820 from March 9, 2012 to May 21, 2012 at 7.5% rate will be

- A. Rs. 22.50
- B. Rs. 27.30
- C. Rs. 28.80
- D. Rs. 29

Ans: B



Permutation & Combination

- What is permutation?
- It is the number of ways a group of things can be arranged.

E.g: Consider 3 letters A,B,C . In how many ways they can be arranged?

- A B C
 - A C B
 - B A C
 - B C A
 - C A B
 - C B A
- 6 ways to arrange these 3 letters

- For 3 letter / 4 letter words its possible but for more number of letters we need a formula-

- $nPr = \frac{n!}{(n-r)!}$



Permutation & Combination

Q. Consider 4 letters A,B,C,D and arrange them in 3 spaces

- - - 3 spaces

No . Of letters = 4

No of spaces = 3

$nPr = 4P_3 = \frac{4!}{(4-3)!} = \frac{4!}{1!} = 4! = 4 \times 3 \times 2 \times 1 = 24$ ways it can be arranged

Q. Arrange 7 letters A,B,C,D,E,F,G in 4 spaces

- - - - 4 spaces

$nPr = 7P_4 = \frac{7!}{(7-4)!} = \frac{7!}{3!} = \frac{5040}{6} = 840$



Permutation & Combination - Remember

$$0! = 1$$

$$1! = 1$$

$$2! = 2 \times 1 = 2$$

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

$$7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$$



Permutation & Combination

Q. In how many ways can the letters of the word 'LEADER' be arranged?

A. 72 B. 144 C. 360 D. 720 E. None of these

Soln:

The word LEADER has 6 letters. So it can be arranged in $6!$ ways.

Out of these 6 letters, 2 letters are repeated (letter E repeated twice)

So we write it as - $\frac{6!}{2!}$

$6!$ ways to arrange letters in the word LEADER

$2!$ In the denominator as letter E is repeated twice

$$= \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1}$$
$$= 360 \text{ ways}$$

Ans : C



Permutation & Combination

Q. Out of 7 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?

- A. 210 B. 1050 C. 25200 D. 21400 E. None of these

Soln:

we need to form a 5 letter word with 3 consonants & 2 vowels = C C C V V

Ways to select, (3 consonants out of 7) AND (2 vowels out of 4)

$= {}^7C_3 \times {}^4C_2 \times 5!$ $\leftarrow$ each group has 5 letters and they can be arranged in 5! ways

$$= \frac{7 \times 6 \times 5}{3 \times 2 \times 1} \times \frac{4 \times 3}{2 \times 1} \times 5!$$

$$= 35 \times 6 \times 120$$

$$= 25200 \text{ ways}$$

Ans : C



Permutation & Combination

Q. From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?

- A. 564 B. 645 C. 735 D. 756 E. None of these

Soln:

We may have (3 men and 2 women) or (4 men and 1 woman) or (5 men only).

Required number of ways = $({}^7C_3 \times {}^6C_2) + ({}^7C_4 \times {}^6C_1) + ({}^7C_5)$

$$= \left(\frac{7 \times 6 \times 5}{3 \times 2 \times 1} \times \frac{6 \times 5}{2 \times 1} \right) + ({}^7C_3 \times {}^6C_1) + ({}^7C_2) \rightarrow [\text{using } {}^nC_r = {}^nC_{(n-r)}]$$

$$= 525 + \left(\frac{7 \times 6 \times 5}{3 \times 2 \times 1} \times \frac{6}{1} \right) + \left(\frac{7 \times 6}{2 \times 1} \right)$$

$$= 525 + 210 + 21$$

$$= 756$$

Ans: D



Difference between permutation and combination

Combination (order does not matter)

"My fruit salad is a combination of apples, grapes and bananas" We don't care what order the fruits are in, they could also be "bananas, grapes and apples" or "grapes, apples and bananas", its the same fruit salad.



Permutation (When the order does matter)

"The combination to the safe is 472". Now we **do** care about the order. "724" won't work, nor will "247". It has to be exactly **4-7-2**.



Difference between permutation and combination

What is permutation?

Permutation: The various ways of arranging a given number of things by taking some or all at a time are all called as permutations.

Permutation includes word formation, number formation, circular permutation, etc. **In permutation, objects are to be arranged in particular order.** It is denoted by ${}^n P_r$ or $P(n, r)$.

Example: Arrange the given 3 numbers 1, 2, 3 by taking two at a time.

Now these numbers can be arranged in 6 different ways: **(12, 21, 13, 31, 23, 32).**

Here,

12 and 21, 13 and 31 or 23 and 32 do not mean the same, because here order of numbers is important.



Difference between permutation and combination

- **What is combination?**

Combination: Each of different groups or selections formed by taking some or all number of objects is called a combination.

Combination is used in different cases which include team/group/committee.

In combination, objects are selected randomly and here order of objects doesn't matter. It is denoted by ${}^n C_r$ or $C(n, r)$ or ${}^n C_r = {}^n C_{(n-r)}$.

Example: If we have to select two girls out of 3 girls X, Y, Z, then find the number of combinations possible.

Now only two girls are to be selected and arranged. Hence, this is possible in 3 different ways: (XY, YZ, XZ,).

Here,
You cannot make a combination as XY and YX, because these combinations mean the same.



Permutation & Combination(Assignment)

Q. In a group of 6 boys and 4 girls, four children are to be selected. In how many different ways can they be selected such that at least one boy should be there?

- A. 159 B. 194 C. 205 D. 209 E. None of these

Soln:

(1 boy and 3 girls) or (2 boys and 2 girls) or (3 boys and 1 girl) or (4 boys).

$$= ({}^6C_1 \times {}^4C_3) + ({}^6C_2 \times {}^4C_2) + ({}^6C_3 \times {}^4C_1) + ({}^6C_4)$$

$$= ({}^6C_1 \times {}^4C_1) + ({}^6C_2 \times {}^4C_2) + ({}^6C_3 \times {}^4C_1) + ({}^6C_2) \rightarrow \text{using } {}^nC_r = {}^nC_{(n-r)} \text{ (to reduce calculation)}$$

$$= (6 \times 4) + \left(\frac{6 \times 5}{2 \times 1} \times \frac{4 \times 3}{2 \times 1} \right) + \left(\frac{6 \times 5 \times 4}{3 \times 2 \times 1} \times 4 \right) + \frac{6 \times 5}{2 \times 1}$$

$$= (24 + 90 + 80 + 15)$$

$$= 209$$

Ans: D



Permutation & Combination(Assignment)

Q. In a group of 6 boys and 4 girls, four children are to be selected. In how many different ways can they be selected such that at least one boy should be there

A. 109

B. 128

C. 138

D. 209

Ans: D



Permutation & Combination(Assignment)

Q. A group consists of 4 girls and 7 boys. In how many ways can a team of 5 members be selected if the team has at least 2 girls?

- A. 322 B. 321 C. 401 D. 116

Ans: A



Permutation & Combination(Assignment)

Q. How many 4-letter words with or without meaning, can be formed out of the letters of the word, 'LOGARITHMS', if repetition of letters is not allowed?

- A. 40
- B. 400
- C. 5040
- D. 2520

Ans: C



Permutation & Combination(Assignment)

Q. In how many different ways can the letters of the word 'MATHEMATICS' be arranged so that the vowels always come together?

- A. 10080
- B. 4989600
- C. 120960
- D. None of these

Ans: C



Permutation & Combination(Assignment)

Q. In how many different ways can the letters of the word 'OPTICAL' be arranged so that the vowels always come together?

- A. 120
- B. 720
- C. 4320
- D. 2160
- E. None of these

Ans: B



Permutation & Combination(Assignment)

Q. How many Permutations of the letters of the word APPLE are there?

A.600 B.120 C.240 D.60

Ans: D



Permutation & Combination(Assignment)

Q. How many different words can be formed using all the letters of the word ALLAHABAD?

A.7560

B.7890

C.7650

D. None of these

Ans: A



Permutation & Combination(Assignment)

Q. Find the value of ${}^{50}P_2$

- A. 4500
- B. 3260
- C. 2450
- D. 1470

Ans : C



Permutation & Combination(Assignment)

Q. How many words can be formed by using letters of the word 'DELHI'?

- a. 50
- b. 72
- c. 85
- d. 120

Ans : D



Permutation & Combination(Assignment)

- Q. Find the number of ways the letters of the word 'RUBBER' can be arranged?
- A. 450
 - B. 362
 - C. 250
 - D. 180

Ans: D



Permutation & Combination(Assignment)

Q. Out of 5 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?

- A. 60
- B. 200
- C. 5230
- D. 7200

Ans : D



Permutation & Combination(Assignment)

Q. In how many ways can a group of 5 men and 2 women be made out of a total of 7 men and 3 women?

- A. 63
- B. 90
- C. 126
- D. 45
- E. 135

Ans: A



Permutation & Combination(Assignment)

Q. In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together?

- A. 360 B. 480 C. 720 D. 5040 E. None of these

Soln:

L E A D I N G $\longrightarrow$ vowels in this word are E,A,I

Remaining letters(consonants) are - L D N G

now we can arrange the vowels together in the remaining spaces as

_ L _ D _ N _ G _ in 5! ways and vowels be rearranged in those spaces in 3! ways

$$5! \times 3! = 720 \text{ ways}$$

Ans : C



Permutation & Combination(Assignment)

Q. In how many different ways can the letters of the word 'CORPORATION' be arranged so that the vowels always come together?

- A. 810 B. 1440 C. 2880 D. 50400 E. 5760

Soln:

C O R P O R A T I O N----- vowels in this word are O,O,A,I,O

Remaining letters(consonants) are - C R P R T N

now we can arrange the vowels together in the remaining spaces as

_C_R_P_R_T_N_ in 7! ways and vowels be rearranged in those spaces in 5! Ways

But the repeated letters are 2R in consonants and 3O in vowels

$$\frac{7!}{2!} \times \frac{5!}{3!} = 50400 \text{ ways}$$

Ans : D



Time & Distance

- **Speed = Distance / Time**
- **Distance = Speed x Time**
- Ram travels from A to B traveling distance of 10 km in 4 hrs. His speed is
- **$10/4 = 2.5 \text{ km/hr}$**
- Ram moves from Pune to Satara at the same speed taking 1 day & 10 hrs. The distance between Pune & Satara is
- **$(24+10) \times 2.5 = 34 \times 2.5 = 85 \text{ km}$**
- Ram now wants to reach back to Pune in 17 hours So he should travel back at a speed of
- **$85/17 = 5 \text{ km/hr}$**



Time & Distance

- If the same distance is traveled at different speeds S_1 & S_2 then average speed is given by-

$$S_a = \frac{(2 \times S_1 \times S_2)}{(S_1 + S_2)}$$

- If the same distance is traveled at different speeds S_1 , S_2 & S_3 then average speed is given by-

$$S_a = \frac{(3 \times S_1 \times S_2 \times S_3)}{(S_1S_2 + S_2S_3 + S_1S_3)}$$

- **Imp : Convert every term to same units**
- **1 Km/hr = $\frac{5}{18}$ m/s & 1 m/s = $\frac{18}{5}$ km/hr**
- If a bowler has a run up of 100 m & he runs at a speed of 36 km/hr the time he takes to complete his runup is
- **$36 \times \frac{5}{18} \text{ m/s} = 10 \text{ m/s}$**
- **$100 \text{ m} \div 10 \text{ m/s} = 10 \text{ s}$**



Time & Distance

If different distance D1,D2 & D3 travelled is at different speeds S1 ,S2 & S3 then average speed is given by-

$$S_a = \frac{(D1 + D2 + D3)}{\left(\frac{D1}{S1} + \frac{D2}{S2} + \frac{D3}{S3}\right)}$$

- Q. A man covers 10kms at a speed of 5 km/hr, 30kms at a speed of 7 km/hr and 20kms at a speed of 15 km/hr. Find out the average speed.
- $S_a = \frac{(10+ 30 + 20)}{\left(\frac{10}{5} + \frac{30}{7} + \frac{20}{15}\right)} = 7.77 \text{ km/hr}$

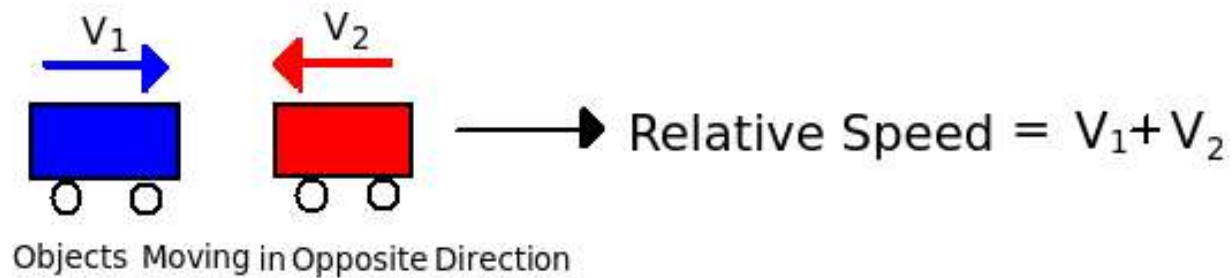
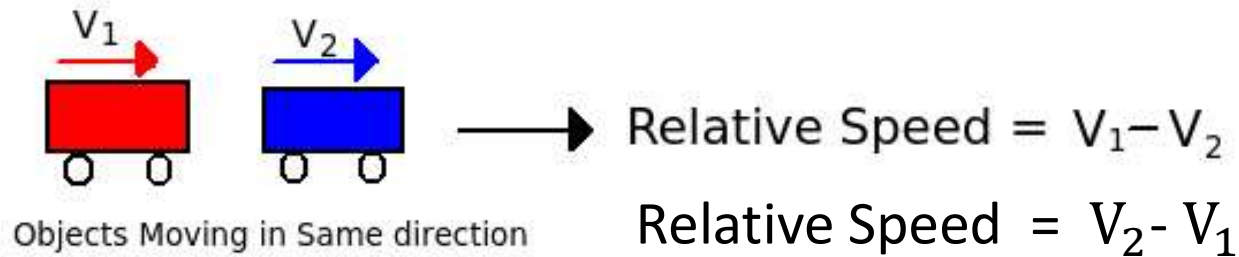


Time & Distance

- Speed & distance are directly proportional.
- $S \propto D$
- Distance & Time are directly proportional.
- $D \propto T$
- Speed & time are inversely proportional.
- $S \propto 1/T$
- **Relative speed** is **defined** as the **speed** of a moving object with respect to another. When two objects are moving in the same direction, **relative speed** is calculated as their difference and if objects are moving in opposite direction then calculate as their sum.
- **Relative speed = $X - Y$ (same direction)**
- **Relative speed = $X + Y$ (opposite direction)**



Relative Speed-



Time & Distance

Q. A car traveled 20% of the time at 40 km/hr, 50% of the time at 20 km/hr and rest of the journey at 30 km/hr. What is the average speed of the car over the whole journey?

- A. 20 km/hr B. 27 km/hr C. 30 km/hr D. 25 km/hr

Soln:

Avg Speed = total dist / total time

Assume Journey = T hr

Total Distance = $(0.2T \times 40 + 0.5T \times 20 + 0.3T \times 30)$
= $8T + 10T + 9T$
= $27T$

Average Speed = $27T/T = 27$ kmph

Ans: B

$$\begin{aligned} S_a &= \frac{(D_1 + D_2 + D_3)}{\left(\frac{D_1}{S_1} + \frac{D_2}{S_2} + \frac{D_3}{S_3}\right)} \\ &= \frac{(20 \times 40 + 50 \times 20 + 30 \times 20)}{\left(\frac{20 \times 40}{40} + \frac{50 \times 20}{20} + \frac{30 \times 30}{30}\right)} \\ &= \frac{2700}{100} = 27 \text{ km/hr} \end{aligned}$$



Time & Distance

Q. At 7:30 am two trains start from their respective stations A & B in opposite direction, 930 km apart at speeds of 60 km/hr & 90 km/hr respectively. At what time do they meet?

A. 12:30 pm B. 1:30 pm C. 1:42 pm D. 1:50 am

Soln:

- Time = Distance/ Speed
- Time = $930 \text{ km} / (60+90) \text{ km/hr}$ (relative Speed adds up)
- Time = 6.20 hours = 6 hrs 12 min
- Time of meeting 1:42 pm

Ans: C



Time & Distance

Q. Walking at a speed of $\frac{4}{5}$ of the original speed a person reaches office 8 min late (8 mins more than normal time). Find the time required usually.

A. 24 min

B. 30 min

C. 32 min

D. 44 min

Soln:

	<u>Original</u>	<u>New</u>
Speed	S	$\frac{4S}{5}$
Time	T	T+8
Speed x Time = Distance is constant		
→ ST	$= \frac{4S}{5} \times (T+8)$	
→ T	$= \frac{4}{5} \times (T+8)$	
→ $\frac{5T}{4} = T+8$		
→ $\frac{5T}{4} - T = 8$		
→ Normal Time T = 32 mins		

Ans: C



Time & Distance

Q. If a train runs at 70 km/hr, it reaches its destination late by 12 minutes. But if it runs at 80 km/hr. it is late by 3 minutes. The correct time needed to cover the journey is?

A. 58 minutes

B. 2 hours

C. 1 hour

D. 59 minutes

Ans: C

Soln:

12mins late

3 mins late

Speed 70

80

Time T+12

T+3

$$70(T+12) = 80(T + 3)$$

$$10T = 600$$

$$T = 60\text{mins}$$



Time & Distance

Two trains start from P and Q respectively and travel towards each other at a speed of 50 km/hr and 40 km/hr respectively. By the time they meet the first train has travelled 100 km more than second train. The distance between P and Q is:

- A. 500 km B. 630 km C. 660 km D. 900 km

Ans: D

At the time of meeting, let the distance travelled by Q, the second train be 'd' km

distance covered by first train $= (d+100)$ km

$$d/40 = (d+100) / 50$$

$$d = 400$$

$$\text{distance between P and Q} = (d+d+100) = (400+400+100) = 900 \text{ km}$$



Time & Distance

Q. A boy rides his bicycle 10km at an average speed of 12km/hr and again travels 12km at an average speed of 10km/hr. His average speed for the entire trip is approximately

- A. 10.4km/hr B. 10.8 km/hr C. 11 km/hr D. 12.2km/hr

Soln:

$$S_a = \frac{(D_1 + D_2)}{\left(\frac{D_1}{S_1} + \frac{D_2}{S_2}\right)}$$

Ans: B



Time & Distance(Assignment)

Q. A boy starts from his house for college at a fixed time. If he walks at the rate of 5 kmph he is late by 7 mins. If he walks at 6 kmph he is 5 min early. Find College to home distance.

A. 5 km

B. 6 km

C. 7 km

D. 6.5 km

	<u>Original</u>	<u>Case1</u>	<u>Case2</u>
Speed	s	5	6
Time	t	t+7	t-5
Speed x Time = Distance is constant			
→	st =	5 x (t+7)/60	= 6 x (t-5)/60
→		5t + 35	= 6t - 30
→		t	= 65 mins
→	Using Case 1 Distance = 5 x (65+7)/60 = 6 km		

Ans B



Time & Distance(Assignment)

Q. One day a person travels to office at $\frac{5}{6}$ of his usual speed. He takes t minutes more than normal time. What is his normal time?

- A. $2t$ B. $3t$ C. $4t$ D. $5t$

Soln:

	<u>Original</u>	<u>New</u>
Speed	S	$\frac{5S}{6}$
Time	T	$T+t$
Speed x Time = Distance is constant		
$\rightarrow ST$	$= \frac{5S}{6} \times (T+t)$	
$\rightarrow T$	$= \frac{5}{6} \times (T+t)$	
$\rightarrow \frac{6T}{5} = T+t$		
$\rightarrow \frac{T}{5} = t \rightarrow \text{Normal Time } T = 5t$		

Ans: D



Time & Distance(Assignment)

Q. A boy goes to school from home at a speed of 10km/hr and return back at 30km/hr. Find his average speed.

A. 15 km/hr

B. 14.5 km/hr

C. 10 km/hr

D. 20 km/hr

Ans: A



Time & Distance(Assignment)

Q. Two trains start from stations A and B and travel towards each other at a speed of 60 kmph and 80 kmph respectively. At the time of their meeting, the second train had travelled 180 km more than the first. The distance between A and B is:

- A. 1260 km
- B. 1620 km
- C. 1660 km
- D. 600 km
- E. 1800 km

Ans: A



Time & Distance(Assignment)

Q. A person travels equal distance with speeds of 3 km/hr, 4 km/hr and 5 km/hr and taken a total time of 47 minutes. The total distance (in km) is :

A. 2 km

B. 3 km

C. 4 km

D. 5 km

Ans: B

If the same distance is traveled at different speeds S_1 , S_2 & S_3 then average speed is given by-

$$S_a = \frac{(3 \times S_1 \times S_2 \times S_3)}{(S_1 S_2 + S_2 S_3 + S_1 S_3)} = \frac{(3 \times 3 \times 4 \times 5)}{(3 \times 4 + 4 \times 5 + 3 \times 5)} = \frac{20 \times 9}{47}$$

Total Dist = Speed x time

$$\begin{aligned} &= \frac{20 \times 9}{47} \times \frac{47}{60} \\ &= 3 \text{ km} \end{aligned}$$



Time & Distance(Assignment)

Q. A man covers half of his journey at 6 km/h and the remaining half at 3 km/h. His average speed is-

A. 9 km/hr

B. 4.5 km/hr

C. 4 km/hr

D. 3 km/hr

Soln:

• Average speed = $\frac{2xy}{x+y} = \frac{2 \times 6 \times 3}{6+3} = \frac{36}{9} = 4 \text{ km/hr}$

Ans: C



Time & Distance(Assignment)

Q. A car traveled 20% of the time at 30 km/hr, 50% of the time at 40 km/hr and rest of the journey at 50 km/hr. What is the average speed of the car over the whole journey?

- A. 40 km/hr B. 35 km/hr C. 41 km/hr D. 45 km/hr

Soln:

Avg Speed = total dist / total time

Assume Journey = T hr

Total Distance = $(0.2T \times 30 + 0.5T \times 40 + 0.3T \times 50)$
 $= 6T + 20T + 15T$
 $= 41T$

Average Speed = $41T/T = 41$ kmph

Ans: C

$$\begin{aligned} S_a &= \frac{(D_1 + D_2 + D_3)}{\left(\frac{D_1}{S_1} + \frac{D_2}{S_2} + \frac{D_3}{S_3}\right)} \\ &= \frac{(20 \times 30 + 50 \times 40 + 30 \times 50)}{\left(\frac{20 \times 30}{30} + \frac{50 \times 40}{40} + \frac{30 \times 50}{50}\right)} \\ &= \frac{4100}{100} = 41 \text{ km/hr} \end{aligned}$$



Time & Distance(Assignment)

Q. On a journey, across Delhi, a Taxi averages 30 kmph for 60% of the distance, 20 kmph for 20% of it and 10kmph for the remainder. The average speed for the whole journey is :

A. 20km/hr

B. 22.5 km/hr

C. 24.625km/hr

D. 25km/hr

Ans: A



Time & Distance(Assignment)

Q. A distance is covered by a cyclist at a certain speed. If a jogger covers half of the distance in double the time, the ratio of the speed of the jogger to that of the cyclist is :

A. 1 : 4

B. 4 : 1

C. 1 : 2

D. 2 : 1

Ans: A



Time & Distance(Assignment)

Q. Walking at a speed of 20% more than the original a person requires 6 min less than normal time. Find the time required usually

A. 24 min

B. 30 min

C. 36 min

D. 44 min

• **Ans C**



Time & Distance(Assignment)

Q. Walking at a speed of 12 km/hr a person reaches 10 min late. But if he walks at 20 km/hr he reaches 14 min early. Find the distance.

A. 9 km

B. 12 km

C. 14 km

D. 15 km

Ans: B



Time & Distance(Assignment)

Q. Two cars started simultaneously travelling toward each other from town A and town B 480km apart. It took first car travelling from town A to town B and car covered the distance in 8hrs and car from town B to town A covers distance in 12hrs. Find distance from town A when they meet?

A. 288km B. 250km C. 380km D. 240km

Ans: A

- Speed of first car = Distance/ time = $480 / 8 = 60\text{km/hr}$
- Speed of second car = Distance/ time = $480 / 12 = 40\text{km/hr}$
- The cars will meet in = $480 / (60+40) = 4.8 \text{ hrs}$ (relative Speed adds up as travelling in opposite directions)
- Dist from A where they will meet = speed of car from A x time
= $60 \times 4.8 = 288\text{km}$



Time & Distance(Assignment)

Q. A car travels $\frac{1}{3}$ of the distance on a straight road with a velocity of 10 km/h, next one-third with a velocity of 20 km/h and the last one-third with a velocity of 60 km/h. Then the average velocity of the car (in km/h) during the whole journey is-

A. 18km/hr

B. 24km/hr

C. 30km/hr

D. 20km/hr

Ans: A

$$\text{Time} = \frac{\text{Dist}}{\text{Speed}}$$

$$\begin{aligned}\text{Total Time} &= \frac{\frac{1}{3}D}{10} + \frac{\frac{1}{3}D}{20} + \frac{\frac{1}{3}D}{60} \\ &= \frac{D}{30} + \frac{D}{60} + \frac{D}{180} \\ &= \frac{6D + 3D + 1D}{180} \\ &= \frac{10D}{180} \text{ hrs}\end{aligned}$$

$$\begin{aligned}\text{Avg velocity} &= \frac{\text{Dist}}{\text{time}} \\ &= \frac{D}{\frac{10D}{180}} \\ &= \frac{180D}{10D} \\ &= 18 \text{ km/hr}\end{aligned}$$



Time & Distance(Assignment)

Q. A man riding his bicycle covers 150 metres in 25 seconds. What is his speed in km per hour ?

- A. 25 km/hr
- B. 21.6 km/hr
- C. 23 km/hr
- D. 20 km/hr

Ans: B



Time & Distance(Assignment)

Q. A motorist travelled the distance between two towns, which is 65 km, in 2 hours and 10 minutes. Find his speed in meter per minute.

- A. 200 meters/min
- B. 500 meters/min
- C. 600 meters/min
- D. 700 meters/min

Ans: B



Trains

- **Trains**

- Let $S1$ = speed of train, $S2$ = Speed of Object
 $L1$ = length of the train, $L2$ = Length of the object.
 t = time taken by train to completely pass the object

Case A : Stationary object without considerable length

$$L1 = S1 \times t$$



Trains

Q. A train running at the speed of 60 km/hr crosses a pole in 9 seconds. What is the length of the train ?

- A. 120 metres B. 180 metres C. 324 metres D. 150 metres

Ans : D

Case A : Stationary object without considerable length

$$\begin{aligned} L_1 &= S_1 \times t \\ &= 60 \times \frac{5}{18} \times 9 \\ &= 150\text{m} \end{aligned}$$



Trains

Q. A train covers 10kms in 10 mins. If it takes 6 secs to pass a pole ,the length of train is?

A. 100m

B. 110m

C. 210m

D. 120m

Ans: A

$$L1 = S1 \times t$$

$$S1 = D/t$$

$$= \frac{10 \times 1000m}{10 \times 60sec}$$

$$= 50/3 \text{ m/sec}$$

$$L1 = S1 \times t$$

$$= \frac{50}{3} \times 6$$

$$= 100m$$



Trains

- **Trains**

- Let $S1$ = speed of train, $S2$ = Speed of Object
 $L1$ = length of the train, $L2$ = Length of the object.
 t = time taken by train to completely pass the object

Case B : Stationary object with considerable length

$$L1 + L2 = S1 \times t$$



Trains

Q. A train of length 600 m crosses a man standing on a platform in 45 sec & the same train crosses the complete platform in 2 min. What is the length of the platform?

A. 500 m B. 700 m C. 900 m D. 1000 m

• **Soln:**

• Case A : $L_1 = S_1 \times t$ (Train passing the man)

$$\begin{aligned} 600 &= S_1 \times 45 \\ S_1 &= 600/45 \\ &= 40/3 \end{aligned}$$

• Case B : $L_1 + L_2 = S_1 \times t$ (Train passing the platform)

$$600 + L_2 = 40/3 \times 120$$

$$L_2 = 1600 - 600$$

$$L_2 = 1000 \text{ m}$$

• **Ans D**



Trains

Q. A train moves past a man and a bridge of 260m long in 8sec and 10sec. What is the speed of train?

A. 140m/s

B. 130m/s

C. 120m/s

D. 110m/s

Ans: B

Soln:

Case 1:

$$L1 = S1 \times t$$

$$L1 = S1 \times 8$$

Case 2:

$$L1 + L2 = S1 \times t$$

$$L1 + 260 = S1 \times 10$$

$$S1 \times 8 + 260 = 10S1$$

$$8S1 + 260 = 10S1$$

$$2S1 = 260$$

$$S1 = 130\text{m/s}$$



Trains

- **Trains**

- Let $S1$ = speed of train, $S2$ = Speed of Object
 $L1$ = length of the train, $L2$ = Length of the object.
 t = time taken by train to completely pass the object

Case C : Moving object without considerable length

$$L1 = (S1 \pm S2) \times t$$



Time & Distance

Q. A train of length 600m running at a speed of 60km/hr crossed a man coming from the opposite direction on a bike in 20 sec. Find the speed of the bike.

A. 24 km/hr

B. 36 km/hr

C. 40 km/hr

D. 48 km/hr

Soln:

$$60 \text{ km/hr} = 60 \times \frac{5}{18} = \frac{50}{3} \text{ m/s}$$

Case B : **$L_1 = (S_t + S_b) \times t$** (Train passing the bike)

$$600 = (\frac{50}{3} + S_b) \times 20$$

$$S_b = \frac{40}{3} \text{ m/s} \times \frac{18}{5} = 48 \text{ km/hr}$$

Ans: D



